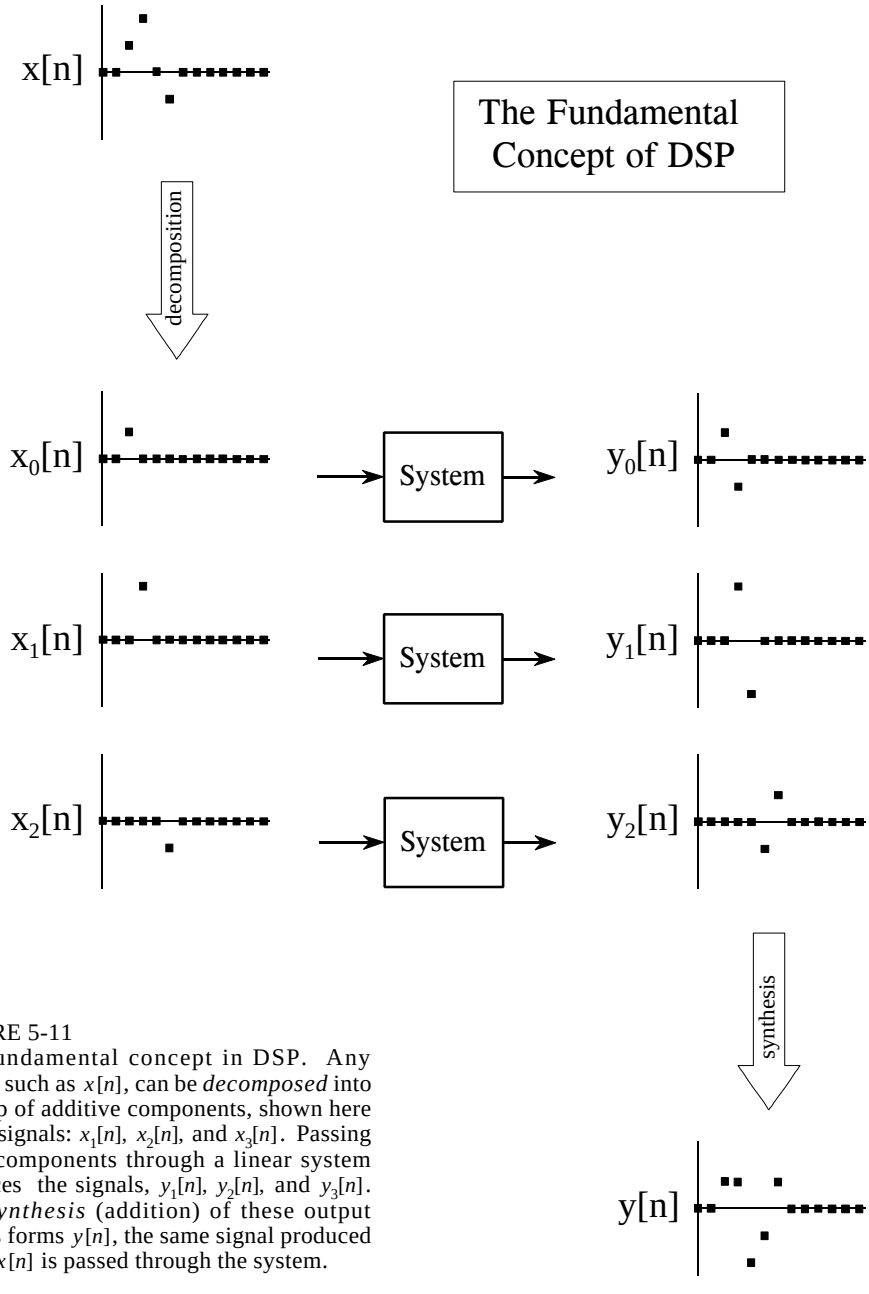

FIGURE 5-11
The fundamental concept in DSP. Any signal, such as $x[n]$, can be *decomposed* into a group of additive components, shown here by the signals: $x_1[n]$, $x_2[n]$, and $x_3[n]$. Passing these components through a linear system produces the signals, $y_1[n]$, $y_2[n]$, and $y_3[n]$. The *synthesis* (addition) of these output signals forms $y[n]$, the same signal produced when $x[n]$ is passed through the system.

signal, called $x[n]$, passing through a linear system, resulting in an output signal, $y[n]$. As illustrated in Fig. 5-11, the input signal can be decomposed into a group of simpler signals: $x_0[n]$, $x_1[n]$, $x_2[n]$, etc. We will call these the **input signal components.** Next, each input signal component is individually passed through the system, resulting in a set of **output signal components**: $y_0[n]$, $y_1[n]$, $y_2[n]$, etc. These output signal components are then synthesized into the output signal, $y[n]$.